

PROJECT • 2026

SDSS • 2026

MalariaSentinel

Spatial Decision Support System

for malaria elimination

From satellite to risk map.

Six stages. One decision.

01	INGESTA	CHIRPS • ERA5 • MERIT DEM • JRC GSW • MODIS • WorldCover
02	SUITABILITY	TWI • habitat suitability • 4 environmental layers
03	ABM	mal-abm-fast • C++20 • xoshiro256** • 7 ops/day
04	DATASET	(state_t + env) → state_{t+1} • monthly pairs
05	U-NET	32→64→128→256→512 • MSE + 0.5•soft-Dice
06	PREDICTION	risk maps • FastAPI • GeoTIFF

02 • THE SYSTEM

Four pieces working together.

01 The satellite *watches*.

Space imagery, climate, water maps. What we already know about the terrain.

| FEEDS

02 The simulation *thinks*.

A program imagines how mosquitoes move. Slow but accurate.

| GENERATES DATA

03 The model *learns*.

Simulation data teaches the model to do it fast, without losing accuracy.

| PRODUCES

04 The prediction *warns*.

A time series, day by day. As far into the future as you want to look.

03 • WHERE WE'RE GOING

You give it a *region*.
It gives you a risk map.

01 Any *country*, any endemic zone. The system adapts.

02 A time series, day by day. You can look a day, a week, or a month ahead.

REGION

04 • WHERE WE ARE

Today we have something. Not what we want yet.

TODAY

A first version that works with real data.

- The simulation is *basic* — only two mosquito life stages.
- The signals are simple: rain and water.
- It only works for one region.

NOW

05 • SUPPORT

This project is made possible by

ANFAIA

Driving Progress with Artificial Intelligence

CULTURE • HEALTH • ROBOTICS • SUSTAINABILITY • ETHICS

github.com/ANFAIA/MalariaSentinel

SDSS • OPEN SOURCE • 2026

Built with the help of agentic AI systems

thanks